

Evaluation of automated-trading strategies using an artificial market

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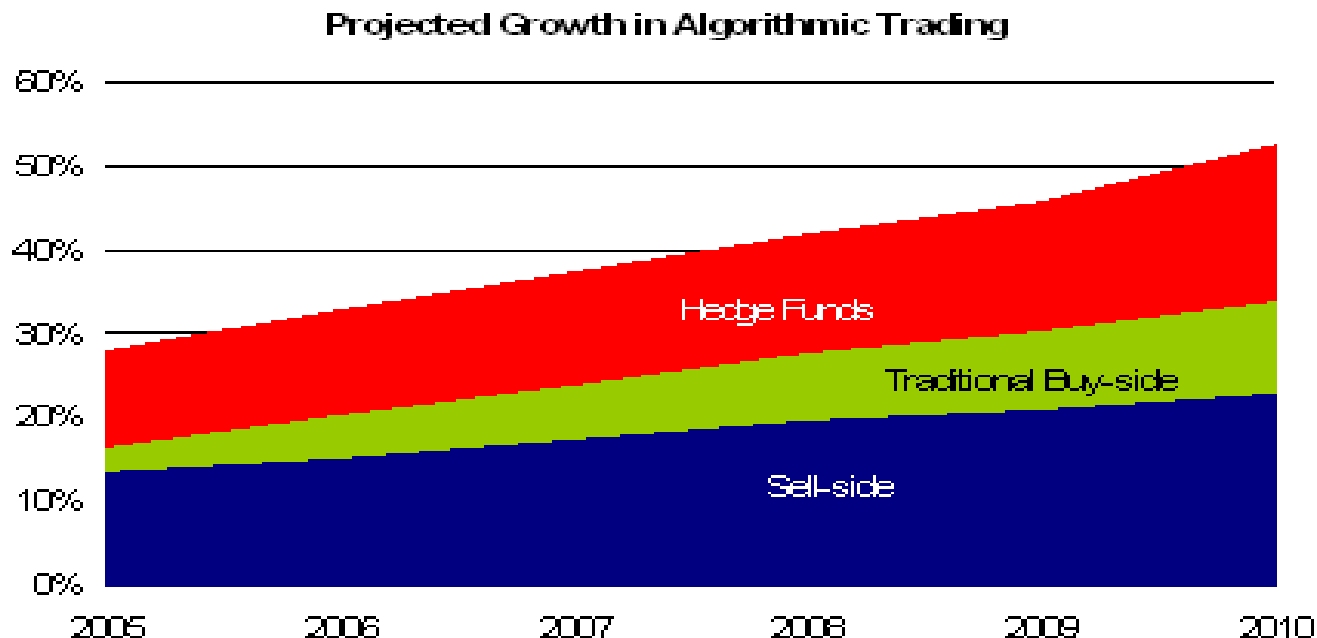
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Abstract

The purpose of this study was to extend the simulation of artificial markets to the practical application. We constructed an artificial-market system with an interface with the automated-trading strategies. Then, using an artificial-market simulation, we conducted two types of evaluations of automated-trading strategies that could participate in a Kaburobo competition. (1) Our system evaluated the risks and returns of the strategies in various market environments. An evaluation using the artificial market was able to provide better information than a conventional evaluation using a back test. (2) Our system could also test the market impact of automated trading strategies. **Our results revealed that the market impact of the strategies may not only depend on their rule but also on the way they are combined with other strategies.**

Introduction

- An **artificial market** is an agent-based model of a financial market.
- Automated trading** is a generic name for the technology of dealing in assets, such as stocks, bonds, and currencies, automatically by computers.
- By the end of 2010, the Aite Group estimates that approximately 53% of all trading in equities will be done through automated trading.



Reference: from Aite Group

Kaburobo competition



Held by the Trade Science Corporation

Rules

- **Initial assets**: Each agent invests 50m yen virtually on the **Tokyo Stock Exchange**. The agents determine their trading behavior automatically and the participants cannot operate nor change their agents during the competition.
- **Stocks**: All stocks included in **Nikkei-225 index and 300 stocks with larger trading volume**.
- **Order method**: The agents must determine **the kinds of stocks, price, and quantity of orders** automatically. They also select a type of orders; **a market order** (市價單) or **a limit order** (限價單). **Short-selling** (放空) is allowed.

- **Timing of orders:**
Morning session: 09:00~11:00 (local time in Japan)
afternoon session 12:30~15:00
- **Order flow:** The orders of agents are sent to the Kaburobo server.
The orders in the contest are virtual, and cannot affect the actual stock prices.
- **Evaluation:** The total profits and losses of the virtual investments are calculated by taking expenses (commissions and interest rates).
Commissions are 0.1% of executed volumes.
Interest rates are charged for short-selling.

Table 1

History of Kaburobo competition.

	2004	2005	2006
Investment period	1 month	3 months	4 months
Number of stocks	40	100	500
Number of participants	2405	2735	6264
Number of agents	3200	4681	8367
Best return (annual rate) (%)	332	828	56.84

Table 2

Average profit (annual rate) of the top 10 programs.

over-learning

	Back test (%)										Out-of-sample test (%)
	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007
Kaburobo	1.2	16.1	44.1	36.6	10.8	11.5	21.5	12.9	17.0	9.5	-9.8
TOPIX	-20.1	-7.5	58.4	-25.5	-19.6	-18.3	23.8	10.2	43.5	1.9	-9.4

System framework

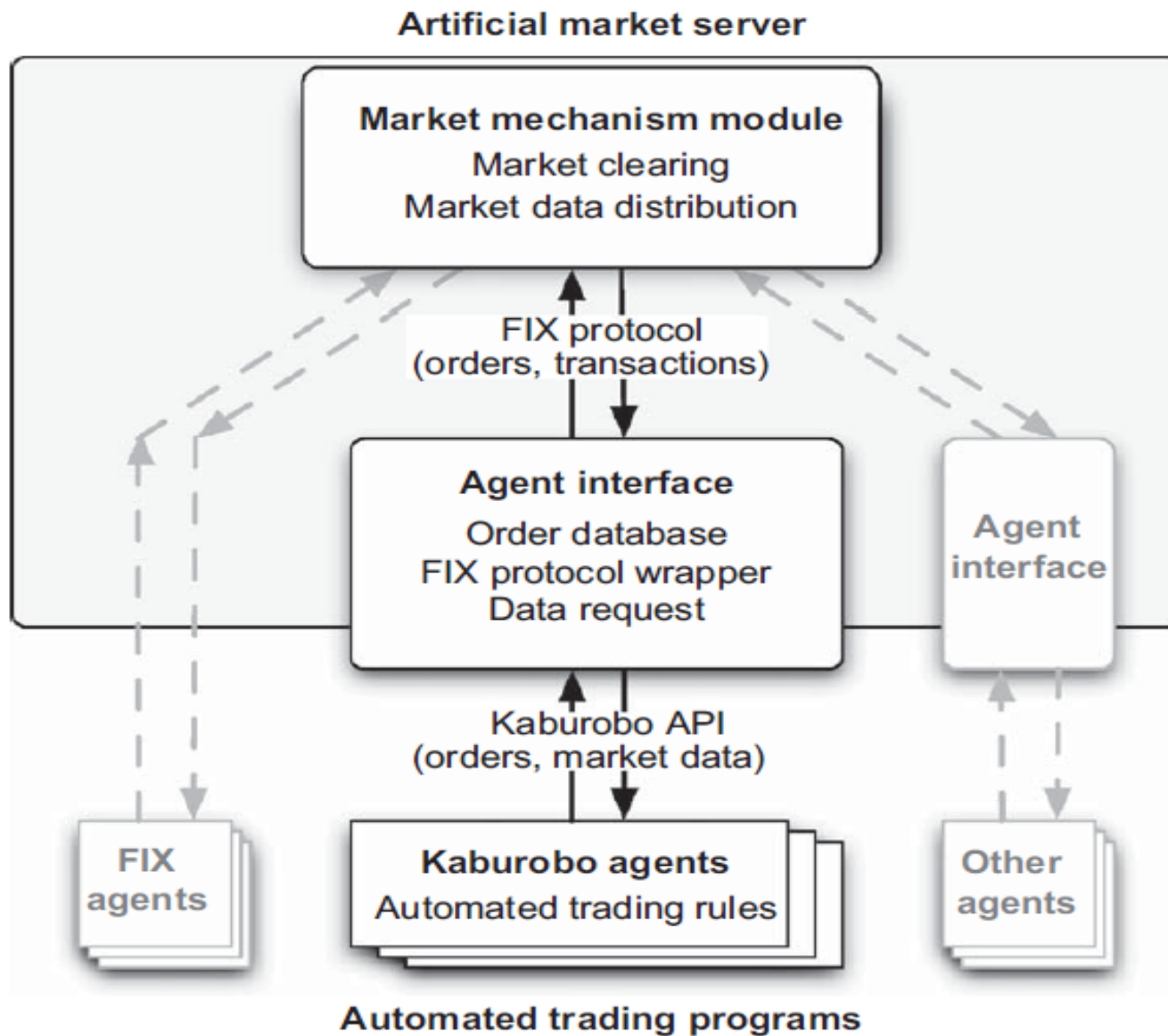


Fig. 1. Framework of our artificial-market system.

Assessment of automated-trading strategies

Table 3

Parameter values of agents

Name	Parameter values
GoldenCross	{SMA, LMA} = {12 day, 24 day}
GoldenCrossShort	{SMA, LMA} = {5 day, 15 day}
GoldenCrossLong	{SMA, LMA} = {25 day, 75 day}
HLBand	$n = 10$
HLBandShort	$n = 5$
HLBandLong	$n = 25$
MACD	{SMA, LMA, PS} = {12 day, 26 day, 9 day}
MACDShort	{SMA, LMA, PS} = {6 day, 12 day, 5 day}
MACDLong	{SMA, LMA, PS} = {16 day, 40 day, 12 day}
Envelope	$P = 0.05$
EnvelopeNarrow	$P = 0.025$
EnvelopeWide	$P = 0.10$
PsychologicalLine	{PH, PL} = {0.75, 0.25}
PsychologicalLineNarrow	{PH, PL} = {0.60, 0.40}
PsychologicalLineWide	{PH, PL} = {0.85, 0.15}
RSI	{RH, RL} = {0.3, 0.7}
RSINarrow	{RH, RL} = {0.2, 0.8}
RSIWide	{RH, RL} = {0.4, 0.6}
TechnicalKun	$n = 20$
TechnicalKunShort	$n = 10$
TechnicalKunLong	$n = 100$
Bayesian	$TH = 0.6$
BayesianNarrow	$TH = 0.4$
BayesianWide	$TH = 0.8$

The details of rules and parameters are written in appendix.

Table 4

Results of back tests in 2005 and 2006.

2005			2006		
	Name	Return		Name	Return
1	GoldenCrossLong	12.92	1	PsychologicalLineWide	15.49
2	BayesianWide	6.35	2	TechnicalKunLong	14.32
3	HLBandShort	3.78	3	Envelope	12.61
4	PsychologicalLineNarrow	3.54	4	RSI	10.09
5	GoldenCrossShort	3.03	5	EnvelopeNarrow	9.71
6	GoldenCross	2.98	6	BayesianNarrow	7.91
7	Bayesian	2.17	7	BayesianWide	7.34
8	HLBand	0.46	8	RSIWide	6.54
9	BayesianNarrow	0.21	9	Bayesian	5.17
10	MACD	0.12	10	RSINarrow	2.89
11	MACDShort	0.08	11	EnvelopeWide	2.01
12	EnvelopeNarrow	0.06	12	GoldenCross	1.92
13	EnvelopeWide	0.00	13	PsychologicalLine	1.89
13	TechnicalKun	0.00	14	TechnicalKunShort	1.87
13	TechnicalKunShort	0.00	15	MACD	1.26
13	TechnicalKunLong	0.00	16	MACDShort	0.87
17	MACDLong	-0.35	17	TechnicalKun	0.08
18	HLBandLong	-0.62	18	PsychologicalLineNarrow	0.05
19	PsychologicalLineWide	-0.89	19	HLBandLong	-0.07
20	PsychologicalLine	-1.15	20	HLBand	-0.79
21	RSINarrow	-1.26	21	GoldenCrossLong	-1.16
22	RSIWide	-1.64	22	MACDLong	-1.23
23	RSI	-4.40	23	HLBandShort	-2.02
24	Envelope	-4.48	24	GoldenCrossShort	-5.64

The correlation coefficient of the two years was -0.462!!.

24 agents with 3 random trading agents.

Orders of agents could change the market price !!

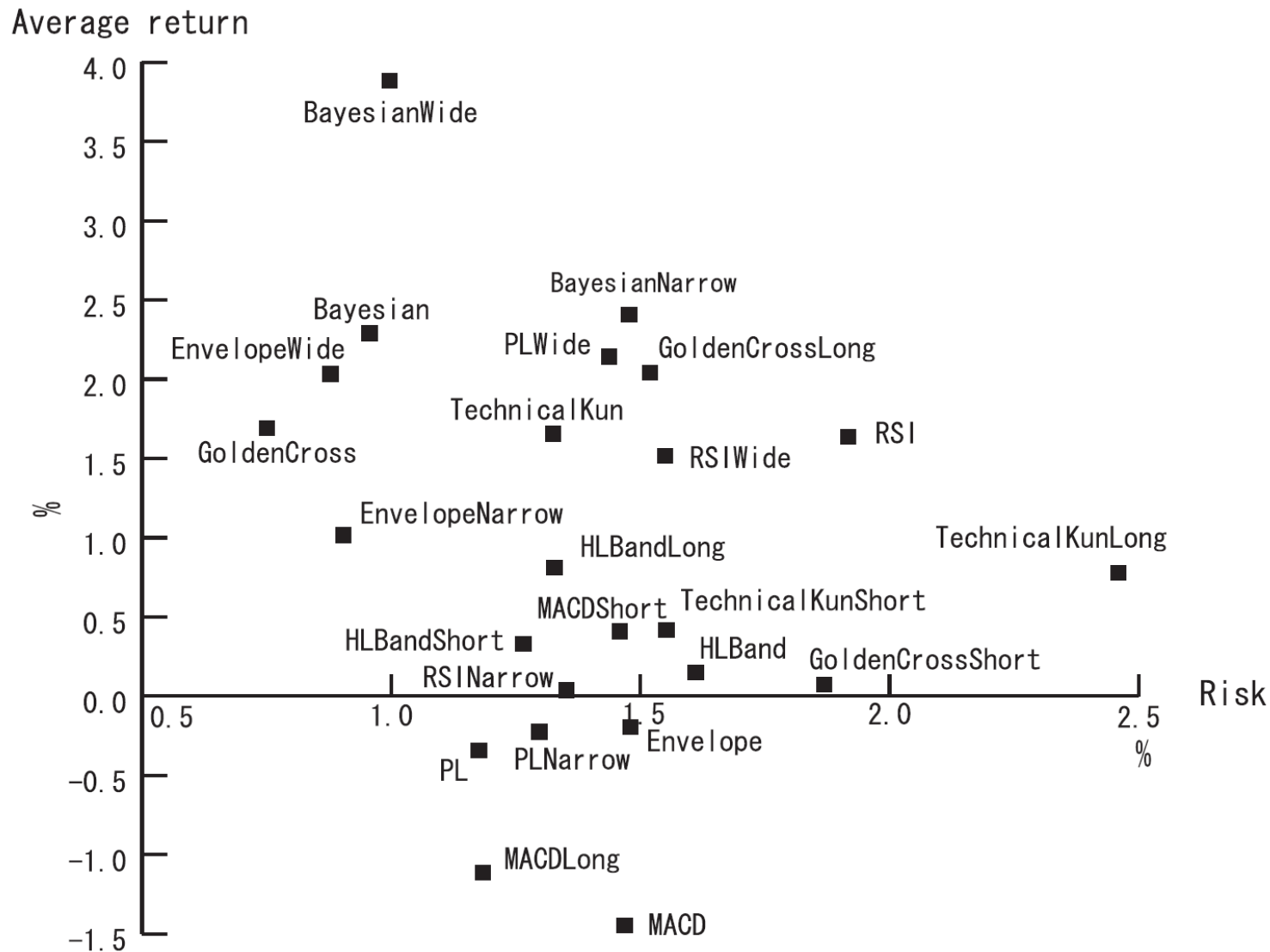


Fig. 2. Plots of risk (standard deviation of returns) and average returns of agents.

Risk: the deviation of final returns over simulation price paths.

Test of impact on market of automated-trading strategies

Test the market impact of two basic trading algorithm:

Momentum trading (trade following trends)

Contrary trading (trade against trends)

Table 5

Agent rules.

Condition		Action
<i>Momentum trading agent</i>		
Upward previous trend	→	Buy 1000 units at $(1 + \alpha_t) \times P_{t-1}$
Downward previous trend	→	Sell 1000 units at $(1 + \alpha_t) \times P_{t-1}$
<i>Contrary trading agent</i>		
Upward previous trend	→	Sell 1000 units at $(1 + \alpha_t) \times P_{t-1}$
Downward previous trend	→	Buy 1000 units at $(1 + \alpha_t) \times P_{t-1}$
<i>Random trading agent</i>		
Probability of $\frac{1}{2}$	→	Buy 1000 units at $(1 + \alpha_t) \times P_{t-1}$
Probability of $\frac{1}{2}$	→	Sell 1000 units at $(1 + \alpha_t) \times P_{t-1}$

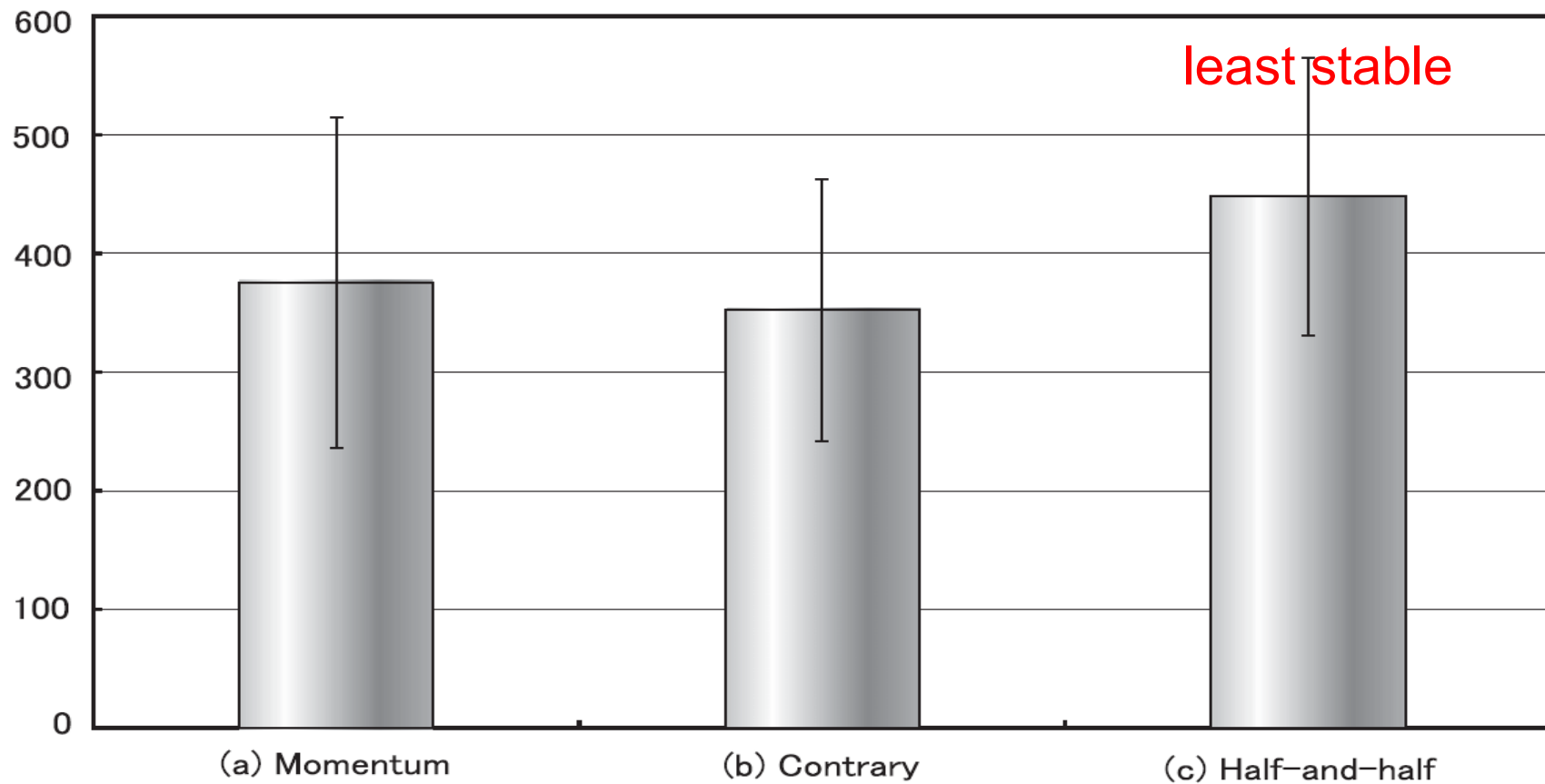
P_{t-1} is a price at the previous session.

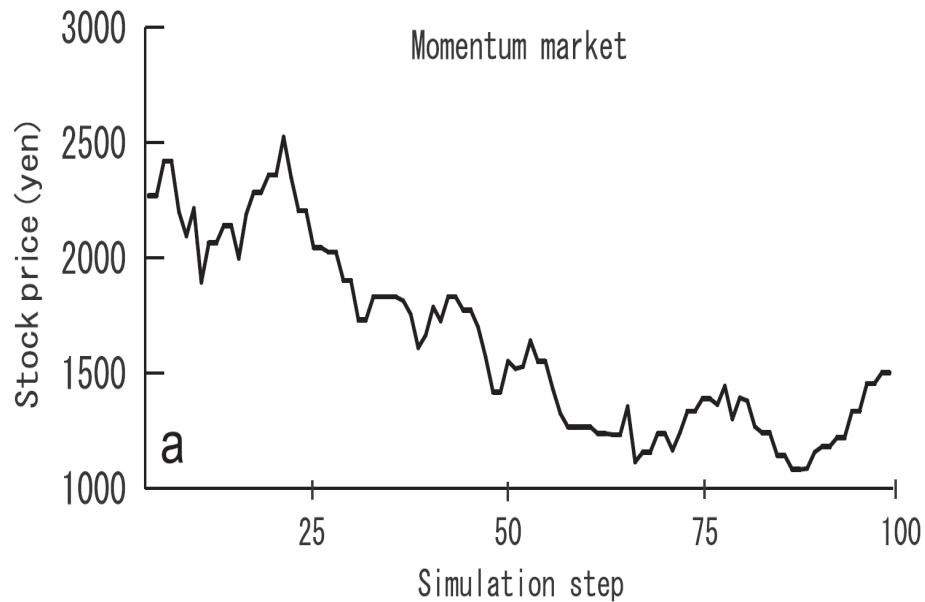
α_t is a random value between -0.1 and 0.1 .

Table 6

Combination of agent types.

	Momentum agent	Contrary agent	Random agent
Momentum market	50	0	5
Contrary market	0	50	5
Half-and-half market	25	25	5

**Fig. 3.** Comparison of market impact: standard deviation of prices (average value over 10 simulation runs).



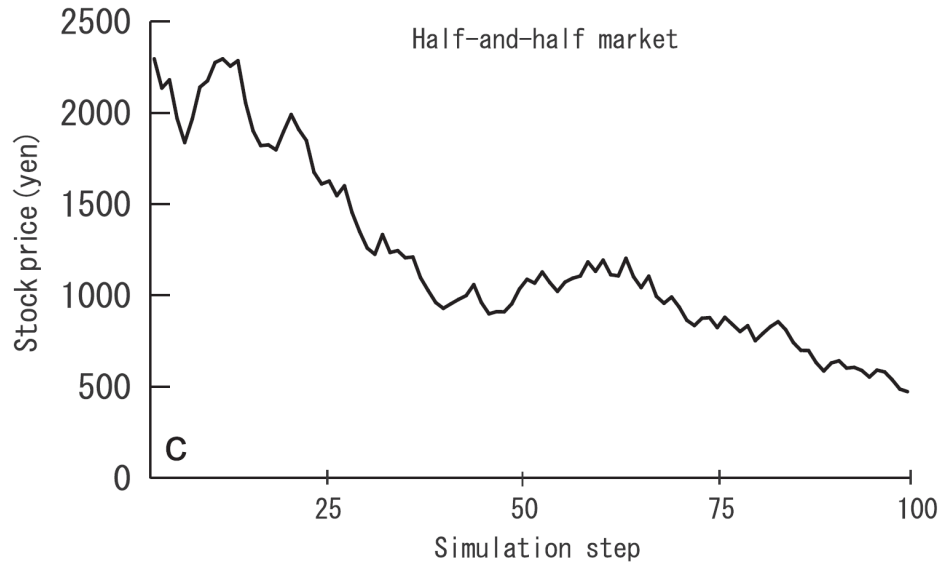
Momentum market:

A short trend in the market. This is because orders by momentum agents supported the trend.



Contrary market:

Almost no trends in the market. This is because orders by contrary agents tended to erase or reverse trends immediately.



Half-and-half market:

A long trend in the market. There were always plenty of orders to both buy and sell sides. However, since orders occasionally to one side in the momentum market, trends did not continue.

Fig. 4. Comparison of price fluctuation of typical paths.

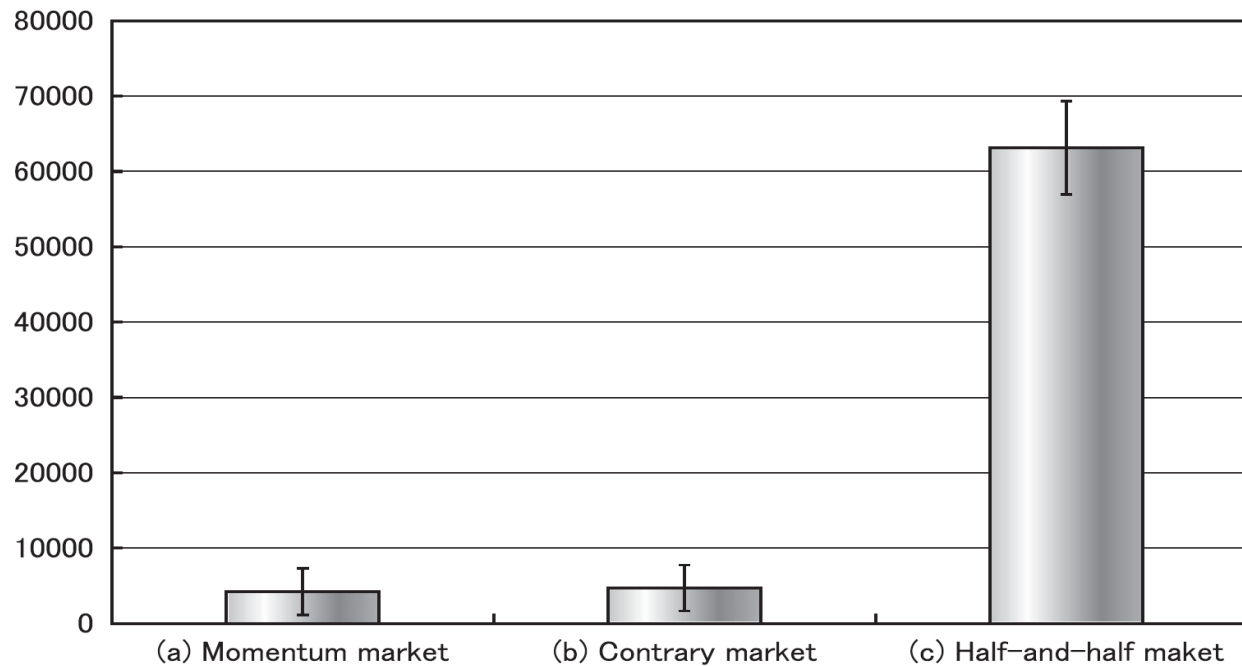


Fig. 5. Comparison of average trading volume of typical paths.

Thank you!!

Japan stock market

日本總共有含東京、大阪、京都、名古屋等在內共八個股票交易所；以東京為主，大阪居次，許多股票都同時在東京及大阪註冊。

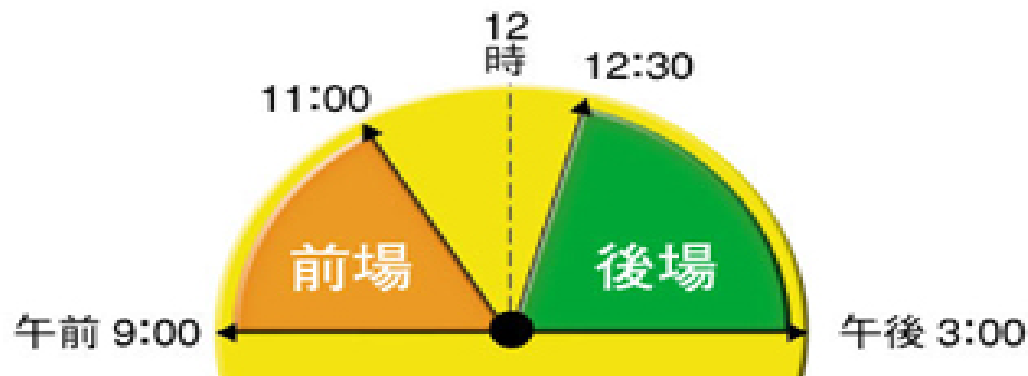
東京證券交易所（全球第二大交易所）分為一部與二部。

東證一部以較大型的公司為主，共 1680 家 (2010/03/31)，股價總市值超過 450 兆日圓，每日成交 5-9 億股以上，規模大、代表性足，所以東證一部指數就成為大部分機構用來評估日本漲跌的指標。

指數方面以日經 225 指數 (NIKKEI) 為代表，該指數是日本經濟新聞社根據日本股市中較具代表性的 225 種股價編制而成，所取樣的股票普遍為中大型股。

- 交易單位：依公司而定，一單位大多為 1000 股或 100 股
- 無任何證券交易稅

取引時間



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